ModelIt!

Student Activity Pack

Where Do Animals Live?

Animal Homes and Habitats

Kindergarten | NGSS K-LS1-1 | Cause and Effect

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 Circle where a duck lives: (picture of a pond / picture of a desert / picture of a house)

2 Draw an animal in its home.

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Amount of Food

Number of Animals

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Little Food

Amount of Food set to low

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Lots of Food

Amount of Food set to high

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Why do different animals live in different places? Use evidence from your simulation to support your answers.

8 STEM Challenge: Build an Animal Habitat

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YOUR ENGINEERING MISSION

Design and build a model habitat for a chosen animal that includes food, water, and shelter.

THE SCENARIO

The city zoo is building new habitats for their animals! Your team has been asked to design the perfect home for one animal. It must have food the animal likes, water to drink, and a safe shelter. Use what you learned about habitats to create the best animal home!

Engineering Design Thinking

1. WHICH ANIMAL DID YOU CHOOSE AND WHERE DOES IT LIVE IN THE WILD?
2. WHAT MATERIALS WILL YOU USE TO SHOW FOOD, WATER, AND SHELTER?
3. HOW WILL YOU SHOW THAT YOUR HABITAT HAS EVERYTHING THE ANIMAL NEEDS?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Zookeepers take care of animals every day — feeding them, keeping their habitats clean, and making sure they are healthy and happy. They earn $28,000-$45,000/year.